

COMP 3270 – Introduction to Algorithms

Homework 4

Topics – Algorithm analysis

1. Find the worst-case time complexity of the MatrixMystery algorithm below.

```
Input: mat:  $n \times n$  array of integers
Input: n: number of rows and columns in mat
1 Algorithm: MatrixMystery
2 for i = 2 to n do
3   for j = 1 to i - 1 do
4     dot = 0
5     for k = 1 to n do
6       dot = dot + mat[k][i] · mat[k][j]
7     end
8     if dot = 0 then
9       return true
10    end
11  end
12 end
13 return false
```

2. Find the worst-case time complexity of the OpenMystery algorithm below.

```
Input: n: positive integer
1 Algorithm: OpenMystery
2 Let open be a Boolean array of size n
3 Initialize open to false
4 nopen = 0
5 for i = 1 to n do
6   j = i
7   while j ≤ n do
8     if open[j] then
9       nopen = nopen - 1
10    else
11      nopen = nopen + 1
12    end
13    open[j] = ¬open[j]
14    j = j + i
15  end
16 end
17 return nopen
```

3. Analyze the worst-case time complexity of the algorithm below.

Input: n : nonnegative integer

1 Algorithm: LoopMystery

2 $sum = 0$

3 $t = 1$

4 $d = 1$

5 $k = n$

6 while $k > 1$ **do**

7 **for** $i = 1$ to k **do**

8 $t = t + d$

9 $sum = sum + t$

10 **end**

11 **if** k is odd **then**

12 $d = -d$

13 **end**

14 $k = \lfloor k/2 \rfloor$

15 end

16 return sum